CommonsLens: Automatic Category Suggestions for Wikimedia Commons

Final Report

TEAM-89

Hackathon Problem Statement 3

1. Problem and Approach

Categories are how files on Wikimedia Commons are found, but many uploads arrive with few or none, and editors must choose from a huge, growing, long-tailed category graph. A fixed classifier fits poorly: the label space changes daily, most categories have a handful of examples, labels are incomplete (a missing category is not a negative), and editors need to know *why* a category is suggested and *when* the system is unsure. CommonsLens treats categories as retrievable objects in a shared image–text space: a frozen vision–language encoder, several complementary scorers, calibrated fusion, hierarchy-aware decoding, and conformal suggestion sets, with evidence shown for every suggestion and an editor always in the loop. Nothing is applied automatically.

2. System

Flow: upload → frozen CLIP ViT-B/32 (SigLIP supported) → scorers on a candidate union → fusion → hierarchy decoding → conformal set or abstention → content type, filters and evidence → editor review.

- **Encoder:** L2-normalised embeddings; uploads add a 0.4-weighted embedding of the central 60% crop, so the main subject outweighs the background.
- **Scorers:** (1) kNN voting over indexed images, softmax(s/τ), editor rejections as negative votes; (2) text prototypes from prompt ensembles; (3) *smoothed prototypes* (text embedding blended with the image centroid, $w{=}n/(n{+}5)$; parent/sibling term tested, Section 4.2); (4) a text-initialised linear head with asymmetric loss for categories with ≥3 images; (5) metadata: title/description embedding, IDF word overlap, year agreement.
- **Fusion:** logistic regression over 17 features on the union of each scorer's top 100, fitted on validation plus copies with metadata removed (30%) or images degraded (30%).
- **Hierarchy decoding:** an ancestor is pushed down when a descendant is confident; the strength is tuned on validation.
- **Conformal suggestion sets:** split conformal calibration on validation, separately for uploads with and without metadata. Members are marked ✓; an empty set means "manual categorisation recommended". α is the smallest value whose median set stays ≤10.
- **Content type first:** Object / Living thing / Scene / Event / Abstract concept, from a classifier on frozen embeddings (939 openly licensed Commons photos, equal share per source category) blended with a vote of the suggestions; "Unsure" below 50%, two types shown when close.
- **Filters:** maintenance and contest categories are never suggested. The display hides bare country names, and over-specific categories (dates, people counts, non-Latin script) unless their score is at least 70%; evaluation scores the model before these display filters.
- **Continual learning:** new categories are suggestible immediately (text prototype, word match), new images are appended to the index, and editor accept/reject feeds back without retraining.
- **Demo:** web UI with evidence images, a public read-only mode and a problem-report form whose reports are de-duplicated and summarised for triage.

3. Data and Protocol

6,303 files from the MediaWiki API: 155 seed categories (107 topical; 48 chosen for balance across continents) plus 1,500 uniformly random files. Hidden and maintenance categories are removed. **Temporal split** by upload time (4,429 / 937 / 937), perceptual-hash and embedding (cos ≥ 0.95) de-duplication. **Realistic validation:** the vocabulary is built from training data (plus seeds and parents) only; validation categories are added after fusion is fitted, giving 24,486 categories at test time; the validation–test F1@5 gap is .014, so validation is a faithful guide. 72% of test labels belong to categories with no training image, and only 36% of test labels are in the vocabulary at all, which bounds recall. Encoder: CLIP ViT-B/32; $K{=}50$, τ tuned on validation.

4. Results

4.1 Main results (937 test images)

	image + metadata		image only	
	M2	**Final**	M2	**Final**
P@1	.350	.348	–	.188
F1@1	.237	.234	–	.118
F1@5	.153	.152	.084	.085
R@10	.330	.336	–	.193
Hierarchy-aware P@5	–	.234	–	.121
mAP (macro)	–	.121	–	–

Accuracy is unchanged from M2 while the system gained conformal sets, the trained content type and smoothed prototypes. About one in three uploads with a title gets a correct category in first place, and 64% of in-vocabulary uploads get one in the top 10. Hierarchy-aware P@5 (.234) is double strict P@5 (.116): many "wrong" suggestions are a parent or child of a true category, which an editor would often accept. The tuned hierarchy decoding leaves F1@5 unchanged (.152 with and without).

Ablation (+meta)

Ranking	F1@5	R@10
Full fusion	**.152**	**.336**
w/o smoothed	.152	.330
Lexical only	.099	.202
Smoothed only	.083	.191
Head only	.070	.172
kNN only	.069	.158
Meta text only	.052	.170
Text proto only	.005	.020

Recall@5 by frequency

Bucket	labels	R
unseen	2412	.003
tail 1–4	318	.200
torso 5–19	217	.439
head 20+	380	.821

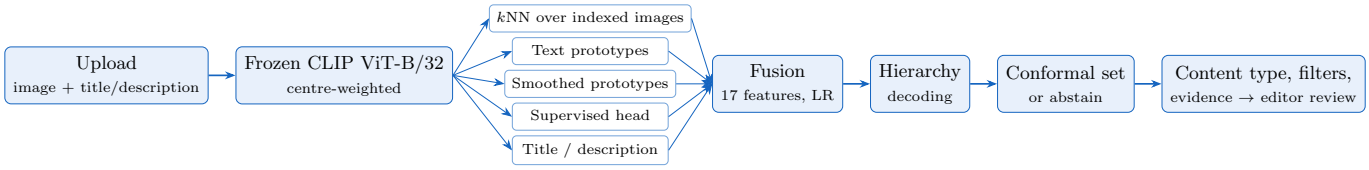


Figure 1: CommonsLens pipeline. Fusion runs only on the union of each scorer's top 100 candidates; editor accept/reject decisions and new categories update the index and vocabulary without retraining.

By category type	Object	Living	Scene	Event	Abstract
Test labels	2151	224	597	106	249
Recall@5	.123	.384	.238	.217	.112
Precision@5	.120	.184	.120	.064	.059

Living things and scenes are easiest; abstract concepts are hardest. Resolution barely matters (F1@5 .155 for ≥2000 px, .143 for 800–2000 px, .144 below 800 px).

4.2 Smoothed prototypes (M1's core idea)

Validation, prototype only			**Test, R@10 by bucket**		
Setting	R@10	tail	Bucket	text	smoothed
Text only	.061	.061	all	.020	**.191**
Text + graph	.049	.053	tail	**.050**	.025
Text + images	**.223**	.061	torso	.055	**.240**
+ graph (γ=.3)	.221	.045	head	.053	**.797**
+ graph (γ=.6)	.213	.030			

Image centroids make the prototype scorer nearly ten times stronger on its own (.020→.191), but the full system does not change because kNN and the head already carry the same evidence. The parent/sibling term, M1's hypothesis for the tail, *lowered* tail recall on validation, so it was not selected: siblings in the Commons graph often look nothing alike. A first version that *replaced* the text prototypes broke title matching, which compares text with text (metadata-text-only F1@5 .052→.007), so the smoothed set is now a separate signal.

4.3 Conformal suggestion sets

With metadata	target	cover.	mean	median	abstain
α=0.3	.70	.624	13.0	12	.009
α=0.4	.60	.518	4.1	3	.069
α=0.5 (default)	.50	.425	2.1	2	.197
Image only, α=0.5	.50	.370	9.6	9	.001
Fixed top-k: top-3 .484	top-5 .556	top-10 .638			

Coverage counts images with a correct category in the set (in-vocabulary images, 98% of the test set). Sets adapt to difficulty: at α=0.4 the median set has 3 categories, beats a fixed top-3 (.518 vs .484), and 7% of images are sent to manual categorisation. Without a title the same guarantee needs sets about five times larger, so the demo encourages adding one. Coverage falls 7–8 points short of the target: test uploads are later than the calibration uploads and meet a larger vocabulary, which breaks the exchangeability the guarantee assumes. Recalibrating on a rolling window of recent uploads (M1's monitoring plan) is required in production.

4.4 Bias audit, robustness, efficiency

F1@5 by group				**Robustness, F1@5**		
Group	n	+meta	img	Degradation	+meta	img
Objects	318	.190	.141	none	.152	.085
Scenes	109	.239	.170	blur r=2	.156	.077
Events	20	.233	.206	blur r=5	.145	.054
Abstract	127	.051	.029	JPEG q10	.151	.074
Europe	42	.274	.037	down ×.15	.152	.063
Africa	71	.207	.006			
S. America	48	.087	.009			
Asia	47	.087	.034			
N. America	43	.036	.012			

Efficiency (laptop, no server GPU): end to end p50 30.1 ms, p95 32.6 ms per image; scoring and ranking 1.0 ms; 5,366 indexed images and 24,486 categories in 58.3 MB of memory (84.3 MB on disk).

4.5 Content type

The classifier is trained on 939 photos (Object 175, Living thing 216, Scene 193, Event 176, Abstract 179) and reaches 82.9% on held-out photos across the five types. On a hand-labelled pack of 250 living subjects (88 of them deliberately hard: partly hidden, old or low-contrast photos), 91.6% were typed Living thing and 88.6% of the hard ones got the right broad type. Portraits were the exception (4 of 14), traced to a collection bug that left the people source categories unused; it is fixed and the classifier retrained. Test-pack photos are never used for training.

5. Findings

- **Fusion is the main win:** .152 vs .099 for the best single scorer (+53%).
- **Metadata is essential for places and robustness:** Africa .207 with a title vs .006 image only; under heavy blur F1@5 drops 5% with metadata but 36% image only.
- **The long tail is the bottleneck:** recall@5 .821 for categories with ≥20 images, .003 for unseen ones, which hold 72% of test labels. Neither text prototypes nor graph smoothing closed this gap.
- **Regional disparity:** with metadata, Europe .274 and Africa .207 but North America .036; image-only accuracy is near zero for every region. Groups are small (42–71 images), so the gaps are indicative only.
- **Evaluation pitfalls we caught:** a validation vocabulary that included validation labels (val–test gap .135→.014 once fixed), and a display-only filter wrongly applied inside the model, which silently zeroed the scores for photos from several regions.

6. Testing

77 automated tests (offline end-to-end train → evaluate → predict → update, conformal coverage on synthetic data, smoothing, collectors, demo). Hands-on testing with our

own photos and tester reports from the public demo led to: centre weighting (people pulled towards "Marathons"), buildings as Scenes, Living thing as a fifth type, mixed types (a fruit market is Object + Scene), hiding dated and non-Latin categories, filtering "Wiki Loves" contest categories, and de-duplicating repeated reports.

7. Responsible Use and Licensing

Only openly licensed Commons files are used for training and indexing, and each keeps its licence. External datasets suggested during development (ObjectNet, WIDER FACE, DTD, COCO) were *not* used for training: their terms forbid training on them, limit use to non-commercial research, or mix such licences, which would conflict with the MIT licence and the sponsor's commercial licence. CommonsLens never identifies people; it only types them as living things. Suggestions are always reviewed by an editor, with the evidence images shown.

8. Editor Experience

Each result starts with a ⋆ row giving the content type (or two, e.g. a fruit market is Object + Scene) and an *Unsure* banner when the type is below 50%. Suggestions follow with their score; rows marked ✓ form the conformal set, and notes explain abstention, a missing title or a low-quality image. For every suggestion the demo shows the indexed images that voted for it and the signals behind its score. Editors can accept or reject suggestions and add new categories; testers can report a problem, which is stored with what the system showed. A public read-only mode hides the editing panels.

9. Deployment and Scalability

The encoder is frozen, so no GPU training is needed and a new deployment only builds an index. Scoring and ranking take 1 ms per image after encoding; the index is a FAISS flat, HNSW or IVF-PQ index chosen by configuration, and candidates keep fusion independent of the number of categories. On Commons the system would run asynchronously on upload events and store suggestions with their evidence for editors; nothing blocks or changes an upload. New categories, new images and editor decisions take effect immediately; the fusion model and conformal thresholds are refitted periodically on recent uploads.

10. Reproducibility

Every number in this report comes from the repository's scripts (`prepare_data.py`, `train.py`, `evaluate.py`, `train_content_type.py`) with a fixed seed; the README lists the commands and links to the built artifacts. All code is MIT-licensed; third-party models, libraries and data are credited in the README.

11. Limitations and Future Work

Only 36% of test labels are reachable, so the next gains are in vocabulary growth and the tail (multilingual category names, geotags, mining sibling categories with editors). Also pending: a SigLIP vs CLIP comparison (supported in the code), evaluation on a test set of the newest uploads (collector built), a human-reviewed score for "wrong" suggestions that are actually valid, and rolling conformal recalibration.